

Week 2 — Convexity and Linear Programming

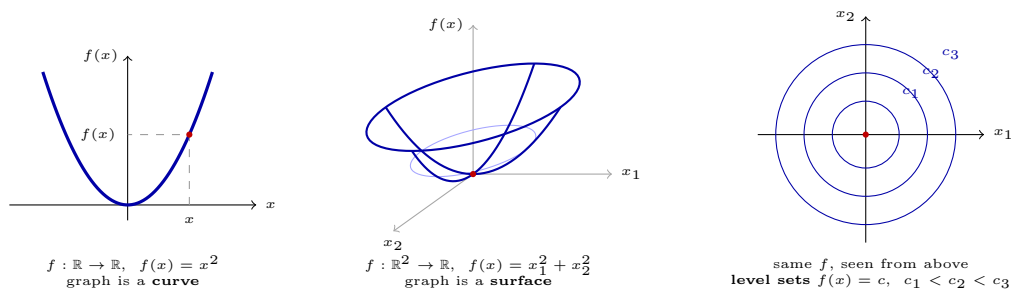
AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Sep 5, 2026

0. Warm-up: functions, vectors, matrices

three objects carry the whole course: a *function*, a *vector*, a *matrix*.

Function $f : A \rightarrow B$ assigns to each input $x \in A$ exactly one output $f(x) \in B$. $A = \text{domain}$, $B = \text{codomain}$.

- $f : \mathbb{R} \rightarrow \mathbb{R}$ one number in, one number out. graph = $\{(x, f(x))\}$ — a curve in the plane
- $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ two numbers in, one number out. graph — a surface in 3D; level set $\{x : f(x) = c\}$ — a curve in the plane
- $f : \mathbb{R}^n \rightarrow \mathbb{R}$ an objective is always of this shape: it must return a *single* number, otherwise two decisions cannot be compared.

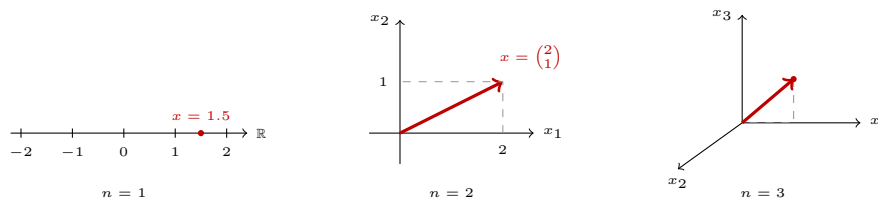


Real space $\mathbb{R}$ = all real numbers — a line. $\mathbb{R}^n$ = lists of n real numbers:

$$x = (x_1, x_2, \dots, x_n), \quad x_i \in \mathbb{R}$$

$n = 1$: line, $n = 2$: plane, $n = 3$: space, $n \geq 4$: same algebra, no picture

“ $x \in \mathbb{R}^n$ ” = one list of n numbers = one point = one arrow from the origin = **one decision**



Vector by convention a *column*: $x \in \mathbb{R}^n$ is $n \times 1$. transpose $x^\top$ is the row, $1 \times n$.

- **inner product** of $c, x \in \mathbb{R}^n$: $c^\top x = c_1 x_1 + \dots + c_n x_n = \sum_{i=1}^n c_i x_i$
as a function: $f(x) = c^\top x, f : \mathbb{R}^n \rightarrow \mathbb{R}$ — fix c , feed it x , get one number
Section 3’s objective: c = prices, x = amounts, $f(x) = c^\top x$ = value of the cargo
- **2-norm** of $x \in \mathbb{R}^n$: $\|x\|_2 = \sqrt{x^\top x} = \sqrt{x_1^2 + \dots + x_n^2}$
as a function: $f(x) = \|x\|_2, f : \mathbb{R}^n \rightarrow \mathbb{R}$ — again one number out

Matrix a matrix is a *stack of row vectors*:

$$A = \begin{pmatrix} - & a_1^\top & - \\ - & a_2^\top & - \\ & \vdots & \\ - & a_m^\top & - \end{pmatrix} \in \mathbb{R}^{m \times n}, \quad a_1, a_2, \dots, a_m \in \mathbb{R}^n$$

m = number of rows, n = length of each row.

Example $m = 3, n = 4$:

$$A = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 3 & 1 & 4 & 0 \\ 0 & 2 & 1 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 4}, \quad a_1 = \begin{pmatrix} 1 \\ 3 \\ 0 \\ 2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 3 \\ 1 \\ 4 \\ 0 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 5 \end{pmatrix} \in \mathbb{R}^4$$

- **matrix $\times$ vector**: for $x \in \mathbb{R}^n$, $Ax \in \mathbb{R}^m$ with $(Ax)_i = a_i^\top x$ — one inner product per row

$$x = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} \in \mathbb{R}^4: \quad Ax = \begin{pmatrix} a_1^\top x \\ a_2^\top x \\ a_3^\top x \end{pmatrix} = \begin{pmatrix} 1 + 6 + 0 + 0 \\ 3 + 2 + 4 + 0 \\ 0 + 4 + 1 + 0 \end{pmatrix} = \begin{pmatrix} 7 \\ 9 \\ 5 \end{pmatrix} \in \mathbb{R}^3$$

as a function: $f(x) = Ax$, $f: \mathbb{R}^4 \rightarrow \mathbb{R}^3$

— takes a 4-dimensional vector, returns a 3-dimensional vector; not a single number (a *scalar*)

in general $f(x) = Ax$, $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

- **identity matrix** $I \in \mathbb{R}^{n \times n}$: 1's on the diagonal, 0's elsewhere. its rows are $e_1^\top, \dots, e_n^\top$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}, \quad (Ix)_i = e_i^\top x = x_i, \text{ so } Ix = x \text{ for every } x$$

- **transpose** $A^\top \in \mathbb{R}^{n \times m}$: rows become columns. A is **symmetric** if $A = A^\top$ (so $m = n$)

Quadratic function

- one variable: $f(x) = px^2 + qx + r$ — a parabola; opens upward iff $p \geq 0$
- n variables: $f(x) = x^\top Qx + q^\top x + r$, $Q \in \mathbb{R}^{n \times n}$ symmetric — still $\mathbb{R}^n \rightarrow \mathbb{R}$, one number out

$$n = 2: \quad x^\top Qx = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{12} & Q_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = Q_{11}x_1^2 + 2Q_{12}x_1x_2 + Q_{22}x_2^2$$

$Q = I$: $x^\top x = x_1^2 + x_2^2$ — the round bowl drawn above. “opens upward” becomes $Q \succeq 0$ (Sections 5, 6).

1. General optimization problem

$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & x \in X \end{array}$$

1. what do we control? **decision variables** x
2. what do we want? **objective function** f
3. what limits us? constraints $\rightarrow$ **feasible set** X

$$X = \{x \in \mathbb{R}^n : g_i(x) \leq 0 \ \forall i \in [m]\}$$

in plain English: x belongs to the n -dimensional real space $\mathbb{R}^n$, **and** it satisfies all m constraints

$$g_1(x) \leq 0, \quad g_2(x) \leq 0, \quad \dots, \quad g_m(x) \leq 0$$

read the symbols: $\in$ = “is in”, $\forall$ = “for all”, $[m] := \{1, \dots, m\}$, s.t. = “subject to”

— so “ $\forall i \in [m]$ ” packs that whole list of m constraints into one stroke

$$h(x) = 0 \iff h \leq 0 \text{ and } -h \leq 0$$

takeaway how we define $f(x)$ and X determines the program class. some problems are easy to solve — but some are not.

2. One problem, three languages

	OR	ML	control
decision	x	θ	u
uncertain data	ξ	[x = input, y = output] state x perturbed by noise w	

keep an eye on that ξ — *uncertain data* is exactly what this course is about. it returns in the week 4 note.

KEY optimization is not a topic inside OR — it is the shared language of OR, ML, control.

watch Brunton, *Optimization: A Bootcamp for Machine Learning, Inverse Problems, and Control* — 22 min, supplementary.

https://www.youtube.com/watch?v=1PBPbGmw1_4

3. Linear programs — and why they are easy

$$\begin{aligned} \max_x \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

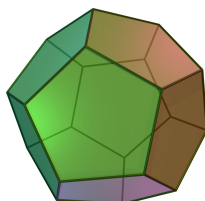
$$c \in \mathbb{R}^n, \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m$$

$$\text{row } i: \quad g_i(x) = a_i^\top x - b_i \leq 0 \iff a_i^\top x \leq b_i$$

$$\text{stack rows} \Rightarrow A \quad \text{so} \quad a_i^\top x \leq b_i \quad \forall i \iff Ax \leq b$$

in many cases variables are nonnegative ($x \geq 0$) — encoded in $Ax \leq b$ by the rows $-x_j \leq 0$

$P = \{x : Ax \leq b\}$ is called a **polyhedron**



Polyhedron in $\mathbb{R}^3$. (image: Wikimedia Commons)

Convexity

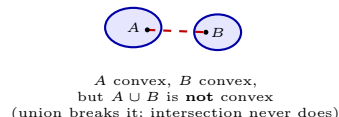
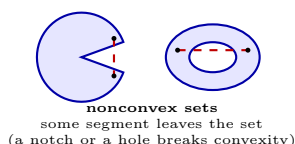
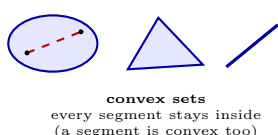
- **Convex set:** $S \subseteq \mathbb{R}^n$ is convex if $\lambda x + (1 - \lambda)y \in S$ for all $x, y \in S$ and all $\lambda \in [0, 1]$
- **Convex function:** $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in \mathbb{R}^n$ and all $\lambda \in [0, 1]$
strictly convex if strict for $x \neq y$, $\lambda \in (0, 1)$
- **Convex problem:** $\min\{f(x) : x \in S\}$ with S convex *and* f convex

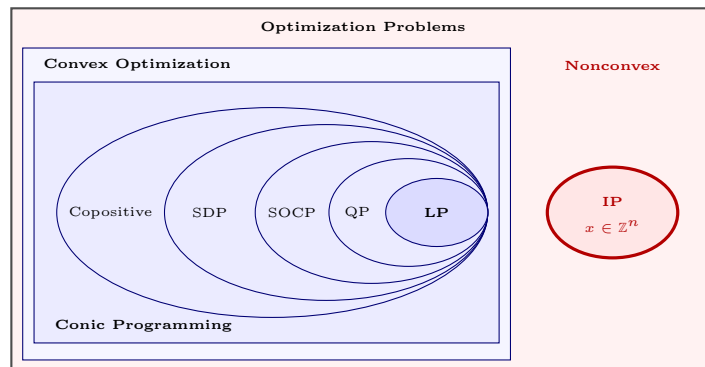
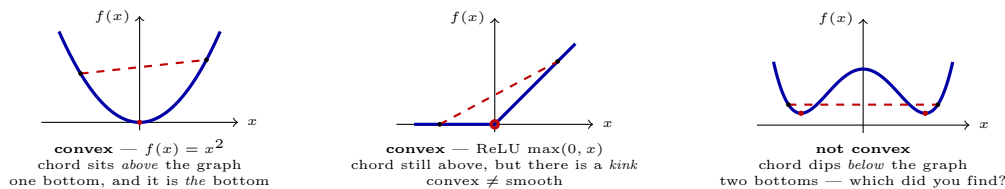
Prop. in a convex problem, every local min is a global min.

pf sketch: if $f(y) < f(x^*)$, then $f(\lambda y + (1 - \lambda)x^*) \leq \lambda f(y) + (1 - \lambda)f(x^*) < f(x^*)$
for all $\lambda \in (0, 1]$ — so better points sit arbitrarily close to x^* .

Examples

convex	not convex
least squares $\ Ax - b\ _2^2$ over unconstrained $\mathcal{X}$ — one bowl	neural-net training loss — many dips
linear $c^\top x$ over P — our LP	$\in \mathbb{Z}$: feasible set turns discrete! — OL7014's turf





LP $\subset$ QP $\subset$ SOCP $\subset$ SDP $\subset$ Copositive $\subset$ Conic $\subset$ Convex, but IP $\not\subset$ Convex
 (QP = quadratic, SOCP = second-order cone, SDP = semidefinite)
 local = global holds everywhere inside the convex box — **this course lives inside it**

takeaway an LP is a convex problem: P convex (intersection of half-spaces) + $c^\top x$ linear, hence a convex function.

4. First LP problem

x = crates of ammo (dense), y = crates of rations (bulky)

$$\begin{aligned} \max \quad & 4x + 5y \\ \text{s.t.} \quad & x + 3y \leq 10 \quad (\text{volume}) \\ & 3x + y \leq 10 \quad (\text{weight}) \\ & x, y \geq 0 \end{aligned}$$

now put it in standard vector form:

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad c = \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 10 \\ 10 \\ 0 \\ 0 \end{pmatrix} \quad (\text{last two rows are } x, y \geq 0)$$

$$\begin{aligned} \max \quad & c^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq b \end{aligned} \quad \text{— Section 3, now with numbers in it}$$

vertices: $(0, 0)$, $(\frac{10}{3}, 0)$, $(2.5, 2.5)$, $(0, \frac{10}{3})$

slide $4x + 5y = z$ up until it leaves $\Rightarrow (x^*, y^*) = (2.5, 2.5), \quad z_{\text{LP}} = 22.5$

KEY 1. an optimum sits at a vertex.

2. fractional answers are normal — nothing forces an LP onto whole numbers.

demo LP-Graphical-Demo.exe — animation slides the line; every vertex labeled with $c^\top v$.

5. One rung up: quadratic programs

one ring out on Section 3's map: keep the polyhedron, let the objective curve.

$$\begin{aligned} \min_x \quad & \frac{1}{2} x^\top Qx + c^\top x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

convex exactly when $Q \succeq 0$ — “the bowl opens upward.” ($Q = 0$: flat bowl, back to LP.)

two QPs we already know: • least squares $\min \|Ax - b\|_2^2$ — a QP with *no* constraints at all • Markowitz (week 4 note, Section 10): $\min x^\top \Sigma x$ s.t. $\mu^\top x \geq r$, $\mathbf{1}^\top x = 1$, $x \geq 0$

Section 4’s cargo, one more request the field unit asks for the exact mix $(\bar{x}, \bar{y}) = (4, 2)$ — infeasible! (weight: $3 \cdot 4 + 2 = 14 > 10$.) so get *as close as possible*:

scalar form

$$\begin{array}{ll} \min & (x - 4)^2 + (y - 2)^2 \\ \text{s.t.} & x + 3y \leq 10 \\ & 3x + y \leq 10 \\ & x, y \geq 0 \end{array}$$

vector form — with numbers in it

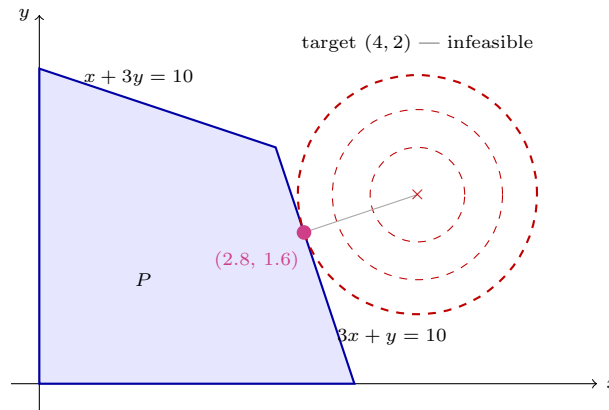
$$\begin{array}{ll} \min & \underbrace{\frac{1}{2} \mathbf{x}^\top \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{x}}_{Q=2I} + \underbrace{\begin{pmatrix} -8 & -4 \end{pmatrix} \mathbf{x}}_{c^\top} + 20 \\ \text{s.t.} & A\mathbf{x} \leq b \quad \text{— same } A, b \text{ as Section 4} \end{array}$$

check: expand $(x - 4)^2 + (y - 2)^2 = x^2 + y^2 - 8x - 4y + 20$; the constant 20 moves no minimizer.

level sets are *circles* around the target ($Q = 2I$: a perfectly round bowl. general $Q \succeq 0$ stretches circles into ellipses.)

solve it graphically: inflate the circle until it first touches P — it lands *tangent to the middle of the weight wall*. drop a perpendicular from the target onto $3x + y = 10$:

$$\boxed{(x^*, y^*) = (2.8, 1.6), \quad f^* = 1.6} \quad \text{— not a vertex.}$$



constrained QP, graphically
inflate the level sets until they first touch P : tangent to an edge — not at a vertex

in the control column of Section 2’s table, the same move — quadratic cost over linear structure — is called *LQR*. same rung, third language.

KEY 1. curved objective $\Rightarrow$ the optimum can leave the vertices: interior (target feasible), edge (here), or vertex.
2. it stops wherever the lowest level set first touches P — LP’s flat level sets slide to a corner.

demo QP-Graphical-Demo.exe — two panels: inflate the circle in 2D, watch the bowl in 3D; drag the target — interior, edge, vertex.

demo week03_markowitz.ipynb — week 3’s Colab — Markowitz as a constrained QP; solve it, then trace the frontier.

read Caron, *Quadratic Programming in Python* — same picture, plus solver code.

<https://scaron.info/blog/quadratic-programming-in-python.html>

takeaway a QP is LP’s polyhedron plus a curved bowl: still convex, still local = global — but the vertex story is over.

6. Convexity-preserving operations

(Boyd & Vandenberghe, *Convex Optimization*, Part I)

$$\text{dom}(f) = \{x \in \mathbb{R}^n : f(x) < +\infty\}$$

Examples of convex functions

<i>univariate</i>	affine function	$ax + b$
	quadratic function	$px^2 + qx + r \quad \text{if } p \geq 0$
	power function	$x^\alpha \quad (\alpha \geq 1 \text{ or } \alpha \leq 0)$
	negative logarithm	$-\log x$
<i>multivariate</i>	affine function	$a^\top x + b$
	quadratic function	$x^\top Px + x^\top q + r \quad \text{if } P \succeq 0$
	norm	$\ x\ _p \quad \text{for } p \geq 1$
$P \succeq 0$ (<i>positive semidefinite</i>) $\iff x^\top Px \geq 0 \quad \forall x \in \mathbb{R}^n$		

Convexity-preserving operations

1) Composition with affine functions:

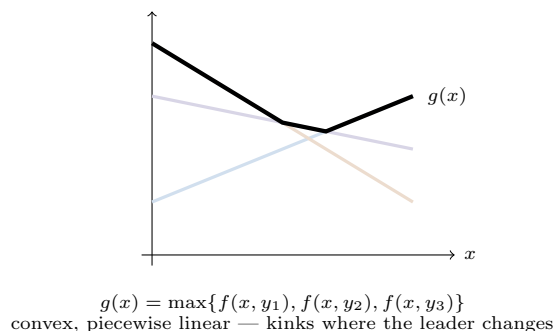
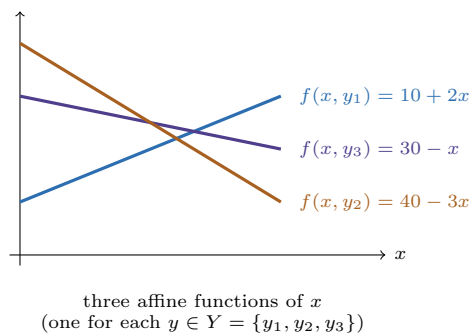
$f(x)$ is convex in x , then $g(x) = f(Ax + b)$ is also convex in x .

2) Nonnegative weighted sum:

$f_1, f_2, \dots, f_k$ are convex in x , then $g(x) = w_1 f_1(x) + \dots + w_k f_k(x)$ is also convex in x for $w_1, w_2, \dots, w_k \geq 0$.

3) Pointwise supremum:

If $f(x, y)$ is convex in x for every fixed $y \in Y$, then $g(x) = \sup_{y \in Y} f(x, y)$ is also convex in x .



4) Partial minimization:

If $f(x, y)$ is *jointly convex* in x and y , and C is a convex set, then $g(x) = \inf \{ f(x, y) : y \in C \}$ is convex in x .

$$\text{jointly convex:} \quad f \left[\lambda \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + (1 - \lambda) \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right] \leq \lambda f \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + (1 - \lambda) f \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

$$\forall \lambda \in [0, 1], \quad \forall \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

$f(x, y) = xy$ is convex in x and in y separately, but *not* jointly convex.

Back to what we have already used

least squares $\|Ax - b\|_2^2$ $\|\cdot\|_2^2$ convex, then (1)

piecewise linear $\max_i (a_i^\top x + b_i)$ affine convex, then (3)

Markowitz $x^\top \Sigma x$ quadratic with $\Sigma \succeq 0$ ($= \|Lx\|_2^2$ for $\Sigma = L^\top L$: (1) again)

polyhedron $P = \{x : Ax \leq b\}$ intersection of m half-spaces — intersection of convex sets is convex

takeaway convexity is compositional: certify the pieces, then check the operation. the risk measures and worst-case objectives ahead are built exactly this way.

next the probability review, and the Markowitz portfolio — see the week 4 note.